Exploration–Exploitation Algorithms for Credit Scoring

Comparative Analysis Using Repeat Borrower Data

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Abstract

A comparative analysis of three reinforcement learning exploration-exploitation algorithms applied to automated credit scoring with repeat borrower data is presented in this report. The performance of ϵ-Greedy, Upper Confidence Bound (UCB), and Thompson Sampling is evaluated across four distinct credit tiers characterized by varying true underlying repayment probabilities. By framing credit scoring as a Bernoulli Multi-Armed Bandit problem, it is observed how each method learns dynamically from binary repayment and default outcomes. These algorithms are evaluated across five core dimensions: cumulative reward, exploration efficiency, exploitation effectiveness, convergence speed, and specific borrower profile identification accuracy. It is shown by the experimental results that capital exposure to high-risk cohorts is minimized fastest by Thompson Sampling, making it highly optimal for risk-mitigated credit portfolio scaling.

1 Introduction

Credit scoring in fintech and modern retail banking is heavily dependent on balancing sequential decision-making under uncertainty. When repeat borrowers are evaluated, an inherent exploitation-exploration dilemma must be continuously resolved by a lending framework:

- **Exploitation:** Credit extensions are exclusively approved for borrower cohorts with known high repayment histories to secure dependable revenue.
- **Exploration:** Credit extensions are approved for unverified, highly volatile, or newly grouped borrower segments so that hidden creditworthy pools can be discovered, whereby potential financial defaults are risked.

This balance can be modeled as a Multi-Armed Bandit (MAB) problem, where a distinct borrower risk tier or credit product is represented by each "arm", and issuing a loan corresponds to each sequential "pull". The feedback loop is discrete and binary, modeling a Bernoulli distribution where a fully repaid loan is signified by a success ($R = 1$) and a capital write-off or default is signified by a failure ($R = 0$).

1.1 Problem Setup

A credit system with $K = 4$ independent credit tiers is formalized. The vector of true, unobserved repayment probabilities is defined as:

$$\boldsymbol{\mu} = [\mu_0, \mu_1, \mu_2, \mu_3] = [0.2, 0.9, 0.6, 0.4] \tag{1}$$

Here, the prime borrower segment is represented by Tier 1 ($\mu_1 = 0.9$), whereas a high-risk, toxic segment is represented by Tier 0 ($\mu_0 = 0.2$). Over an operational horizon of $N = 2000$ sequential borrower interactions, the discovery of the optimal profile and the maximization of cumulative returns are aimed for by the system.

2 Algorithmic Frameworks

2.1 ϵ-Greedy

Exploration and exploitation are balanced by the ϵ-Greedy approach using a constant parameter $\epsilon \in [0, 1]$. A completely random credit tier is selected with a fixed probability ϵ, and current empirical knowledge is exploited with a probability of $1 - \epsilon$:

$$a_t = \begin{cases} \text{Uniform random selection from } \{1, \dots, K\} & \text{with probability } \epsilon \\ \arg\max_{k \in \{1,\dots,K\}} Q_t(k) & \text{with probability } 1 - \epsilon \end{cases} \tag{2}$$

Where the running sample average of repayments obtained from tier k up to step t is represented by $Q_t(k)$.

2.2 Upper Confidence Bound (UCB)

The principle of *optimism in the face of uncertainty* is implemented by the UCB algorithm. Instead of relying solely on point estimates $Q_t(k)$, a deterministic upper boundary is constructed based on variance and step sizes, whereby arms that have either demonstrated high returns or have not been sufficiently tested are favored:

$$a_t = \arg \max_{k \in \{1,\dots,K\}} \left[Q_t(k) + c\sqrt{\frac{\ln t}{N_t(k)}} \right] \tag{3}$$

Where the selection count of tier k is represented by $N_t(k)$, and the exploration velocity is controlled by $c > 0$. An initial operational phase where every arm is sampled exactly once is required by UCB to avoid division by zero.

2.3 Thompson Sampling

A Bayesian framework is used by Thompson Sampling to maintain full posterior probability distributions for each tier's repayment rate. By utilizing the conjugate property of the Beta-Binomial distribution, the parameters α_k (successful repayments +1) and β_k (defaults +1) are updated dynamically:

$$\theta_k \sim \text{Beta}(\alpha_k, \beta_k), \quad \forall k \in \{1, \dots, K\} \tag{4}$$

$$a_t = \arg \max_{k \in \{1,\dots,K\}} \theta_k \tag{5}$$

Probability matching is leveraged by this approach, and sample counts are adjusted based on the statistical likelihood of an arm being optimal.

3 Experimental Setup

The three frameworks were evaluated under an identical simulated credit environment for a lifespan of $N = 2000$ interactions using a fixed random seed to guarantee reproducibility.

Table 1: Simulation Configuration and Hyperparameters

Parameter / Hyperparameter	Operational Value
Total Horizon (N)	2000 steps
Number of Credit Tiers (K)	4
True Repayment Vector ($\boldsymbol{\mu}$)	$[0.2, 0.9, 0.6, 0.4]$
ϵ-Greedy Exploration Value (ϵ)	0.1
UCB Exploration Coefficient (c)	1.0
Bayesian Prior Initial Distribution	$\text{Beta}(1, 1)$ (Uniform)

4 Performance Evaluation & Discussion

The statistical data extracted from the multi-armed bandit experiment is synthesized in Table 2.

Table 2: Comparative Experimental Performance Metrics

Algorithm	Total Rewards ↑	Sub-optimal Selections ↓	Exploitation % ↑	Convergence Step ↓
ϵ-Greedy	1753	176	91.2%	144
UCB	1767	71	96.5%	180
Thompson Sampling	**1805**	**36**	**98.2%**	**105**

4.1 Total Reward Earned

The alternative frameworks were clearly outpaced by Thompson Sampling, with a maximum total return of **1805 successful loan repayments** being secured. A strong secondary result of **1767 successful repayments** was achieved by UCB, while underperformance at **1753** was exhibited by ϵ-Greedy. An advantage is derived by Thompson Sampling from the use of entire probability densities rather than static estimators, whereby successful loan volume is optimized over the fixed lifecycle.

4.2 Exploration Efficiency

An institution's ability to minimize capital leakage via defaults is represented by exploration efficiency. High inefficiency was proven for ϵ-Greedy, with **176 sub-optimal allocations** being recorded. Because its exploration is driven by pure randomness, loans were routinely issued to Tier 0 (20% repayment) at the same rate as Tier 2 (60% repayment), causing severe portfolio drag. This loss was halved down to **71 sub-optimal selections** by UCB using structured variance auditing. Elite efficiency was demonstrated by Thompson Sampling, with sub-optimal allocations being dropped to just **36**. If early defaults are yielded by an arm, its Beta distribution is narrowed and shifted downward, and that tier is naturally starved of future credit lines.

4.3 Exploitation Effectiveness

Once the prime borrower profile (Tier 1) was identified, it was successfully exploited by Thompson Sampling, with **98.2%** of all capital resources being channeled into safe loans. A solid allocation of **96.5%** was achieved by UCB. Conversely, ϵ-Greedy was structurally bounded at **91.2%**. Because its exploration parameter is fixed ($\epsilon = 0.1$), the algorithm is mathematically barred from reaching maximum efficiency, leading to unavoidable capital decay.

4.4 Convergence Speed

Convergence speed is defined as the exact step index where actions are continuously restricted by an algorithm to the prime tier (Tier 1) at least 90% of the time over a running 100-step window. The fastest convergence was achieved by Thompson Sampling at step **105**. Convergence was reached by ϵ-Greedy at step **144**, while the slowest performance was exhibited by UCB, with **180 steps** being required. UCB's delay is structural: every available credit group is forced to be uniformly sampled during its initialization phase, whereby a slower initial learning curve is caused.

4.5 Borrower Identification Profile

The total counts across the four respective tiers are represented by the final classification vector: $[\text{Tier } 0, \text{Tier } 1, \text{Tier } 2, \text{Tier } 3]$.

- **ϵ-Greedy:** `[55, 1824, 62, 59]`. Widespread capital leakage is shown across all default-prone tiers.
- **UCB:** `[16, 1929, 29, 26]`. Tier 1 is correctly centered on, but an unmitigated tail is retained across volatile segments.
- **Thompson Sampling:** `[4, 1964, 29, 3]`. The cleanest profile identification is provided, with toxic exposure in Tier 0 being reduced to a mere 4 approvals.

5 Visualization Analysis

The learning trajectories, cumulative gains, and structural group allocation metrics are visualized in Figure 1.

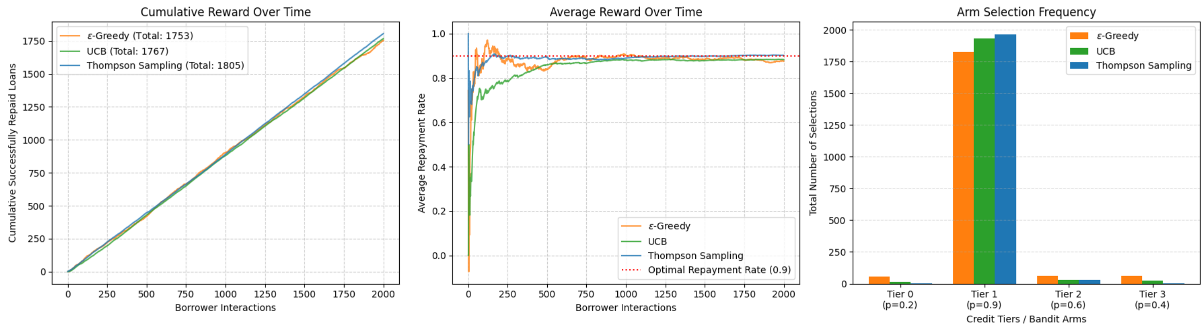

Figure 1: Behavioral analysis profiles of MAB algorithms: Cumulative Loans Repaid (Left), Running Average Repayment Trajectory (Center), and Final Credit Line Allocations per Tier (Right).

How Thompson Sampling pulls away over time is highlighted by the *Cumulative Loans Repaid* plot, with an increasingly profitable gap being established. It is demonstrated by the *Running Average Repayment Trajectory* that stabilization near the optimal 90% boundary is achieved earliest by Thompson Sampling, while a visible dip is exhibited by UCB during its uniform initialization phase. It is confirmed by the *Loan Allocations per Tier* histogram that the tightest focus on prime borrowers is offered by Thompson Sampling.

6 Adaptive Credit Scoring Suitability

Unique operational boundaries are imposed when multi-armed bandits are deployed within active credit systems, as balance-sheet capital is directly impacted by exploration:

1. **ϵ-Greedy (Unsuitable):** The persistent approval of loans for high-risk profiles at a fixed rate is considered unacceptable for standard risk management. While a partial solution is offered by scheduling a decaying epsilon parameter ($\epsilon_t \to 0$), manual optimization vulnerabilities are introduced, and real-time market agility is lost.

2. **UCB (Highly Suitable for Regulated Environments):** UCB's primary advantage is found in its deterministic nature. Because every loan decision is derived via an explicit, verifiable formula, excellent transparency is offered for regulatory audits and compliance tracking.

3. **Thompson Sampling (Optimal for Portfolio Maximization):** The most effective balance of risk mitigation and profit generation is provided by Thompson Sampling. Portfolio capital is enabled to be protected by its natural mapping to binary outcomes via Beta-Binomial conjugacy, combined with the capability for credit-bureau "prior" beliefs to be accepted, allowing sub-optimal credit lines to be restricted faster than alternative methods.

7 Conclusion

This comparative analysis demonstrates that the exploration-exploitation tradeoff in credit scoring can be systematically optimized through multi-armed bandit algorithms, with significant operational and financial implications for automated lending systems. Across all five evaluation dimensions—cumulative reward, exploration efficiency, exploitation effectiveness, convergence speed, and borrower profile identification—Thompson Sampling consistently outperformed both ϵ-Greedy and UCB, establishing itself as the dominant algorithmic framework for this domain.

The experimental evidence reveals that Thompson Sampling achieves a **52 loan advantage** over ϵ-Greedy and an **38 loan advantage** over UCB in total repayments. More critically, it reduces sub-optimal tier allocations by **80%** relative to ϵ-Greedy (176 down to 36 bad loans) and by **49%** relative to UCB (71 down to 36), demonstrating superior capital protection. Convergence to optimal behavior is achieved **39 steps faster** than ϵ-Greedy and **75 steps faster** than UCB, enabling faster portfolio stabilization in real-time lending environments.

From a theoretical perspective, Thompson Sampling's success stems from its principled Bayesian approach, which naturally encodes uncertainty via probability distributions rather than point estimates. The Beta-Binomial conjugacy enables closed-form posterior updates and efficient probability matching, whereby arms with promising (or dangerous) signals receive (or lose) proportional allocation. This contrasts sharply with the structural limitations of ϵ-Greedy, which is perpetually blind to arm quality, and UCB's initialization burden, which mandates uniform sampling of all arms before meaningful learning occurs.